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Reduction in liquid fragility and enhancement in ionic conductivity in the glassy phase of [BmIm][NTf₂] ionic liquid under nano-confinement: PALS and BDS investigations

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Positron annihilation lifetime spectroscopy (PALS) and broadband dielectric spectroscopy (BDS) have evidenced fundamental changes in the structural and dynamic properties of the room temperature ionic liquid (RTIL or IL) 1-butyl-3-methylimidazolium-bis(trifluoromethylsulfonyl)-imide ([BmIm][NTf₂]) under confinement in SBA-15 nano-pores (average pore size of ~5 nm). It is observed that the nano-confinement transformed the fragility of the IL [BmIm][NTf₂] from high to low, resulting in a reduction in the ionic conductivity and diffusion coefficient of the ions inside the nano-pores at room temperature. However, compared with the bulk, in the glassy phase (below 200 K), an increase in both the ionic conductivity and diffusion coefficient is observed, which is due to enhanced molecular free volume under nano-confinement below 200 K. The results of this study are relevant for the fundamental understanding of the intriguing properties of [BmIm][NTf₂] (IL) confined inside nano-pores, which can be employed for several important spatial confinement applications.

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1. Introduction

Ionic liquids (ILs), generally consisting of organic cations and weakly coordinating organic or inorganic anions, are a class of salts having unique properties like low melting points, low vapor pressures, wide liquid ranges, high thermal stability, tunability, high electrical conductivity and wide electrochemical windows.^{1–3} Owing to these unique properties, ILs are considered suitable candidates for several applications, including in energy storage devices as electrolytes,^{4–8} in tribology as lubricants,^{9,10} and in chemical industries as separation media and solvents.^{11,12} In these applications, ILs mostly work under spatial confinement; hence, the most relevant processes in determining a device's performance occur at the liquid/solid interfaces. The properties of phase transition and ion mobility of ILs under nano-confinement significantly differ from their bulk phase due to the combined effect of reduced dimensionality and pore wall interactions with the ILs.^{13–20} Experimental and theoretical studies have shown both faster^{17,18} and slower

dynamics^{19,20} of ILs in nano-confinement depending on the pore size and the physico-chemical nature of the pore walls and ILs. Iacob *et al.*, through combined studies using broadband dielectric spectroscopy (BDS) and pulsed field gradient nuclear magnetic resonance (PFG-NMR) spectroscopy on ILs confined in unidirectional silica nanoporous (~7.5 nm pore size) membranes, have shown an enhanced diffusivity of ILs compared to that of bulk ILs.¹⁷ A similar enhancement in the diffusion coefficient was reported by Shi *et al.* in their theoretical study using molecular dynamics in combination with Monte-Carlo simulations on ILs confined in carbon nanotubes (~2.6 nm pore size).¹⁸ The enhancement in ionic conductivity inside nano-pores was explained based on the lower mass density of ILs due to a reduced packing fraction resulting in reduced interactions among the IL molecules. On the other hand, Busch *et al.* used quasi-elastic neutron scattering (QENS) to study ILs confined in nanoporous carbon and reported a reduced self-diffusion coefficient of ILs due to a hierarchical pore structure with a pore size distribution (~0.6–10 nm) featuring the coexistence of micro- and meso-pores.¹⁹ Le Bideau *et al.* through NMR study on ILs confined in monolithic silica, have also reported slower diffusion of ILs as compared to that in bulk ILs.²⁰ The slowdown of the dynamics was related to the reduced pore size. Similarly, Davenport *et al.* reported almost three times reduction in the conductance of ILs in small single-pores (<10 nm) of a polyethyleneterephthalate film, which was

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attributed to the electrostatic interaction of ILs with the nanopore walls due to the squeezing of ILs inside the small pores.²¹ Several research articles have been published over the past two decades on the changes in the molecular mobility of ILs inside nano-pores. Studies are still in progress aiming at acquiring coherent knowledge on molecular dynamics responsible for the reduction and enhancement in self-diffusivity and ionic conduction of ILs inside the confinement. [BmIm][NTf₂] is a well-studied ionic liquid in its bulk phase, and it has applications in several areas, like electrochemistry, biomass processing, gas separation and adsorption.²² It has the ability to form solid layer-like structures or liquid-like domains on solid surfaces based on the substrate structure.²³ This unique behavior urges us to understand the structural and dynamic properties of the liquid–solid interface of [BmIm][NTf₂] in its confined state.

In the present work, we have used positron annihilation lifetime spectroscopy (PALS) to probe the molecular free-volume of ILs in nano-confinement to understand its effect on altering self-diffusion properties of nano-confined ILs. PALS is a unique *in situ* technique that is used to study the molecular free volume of the ILs in bulk and confined state through measurement of the positronium lifetime and intensity inside the molecular free volume of the matrix.^{24–27} A positron, the *anti*-particle of an electron, can form a bound state with an electron known as a positronium (Ps) in liquid or solid dielectric soft media like polymers and porous powders. The Ps atom energetically tends to be localized in the pre-existing free-volume, or it can also create by itself a cavity/bubble in a liquid with low surface tension by virtue of exchange repulsion with the surrounding electrons, becoming trapped in it.²⁸ Ps may exist in two spin states, depending on the relative spin orientation of the electron and positron: the spin-triplet state or *ortho*-Ps (*o*-Ps), which annihilates in a three-photon mode with an average lifetime of 142 ns in vacuum, and the spin-singlet state or *para*-Ps (*p*-Ps), which annihilates in a two-photon mode with a short average lifetime of 0.125 ns in vacuum. In a molecular medium, *o*-Ps may be localized inside a free volume or pore, and its lifetime may be quenched to around 1–10 ns by seeking out an electron with the opposite spin from the inner surface of the free-volume wall and annihilating in a two-photon mode through an additional pathway, known as pick-off quenching. The *o*-Ps pick-off quenching rate depends on the size and the surface properties of the free volume in the medium. Thus, the changes in free-volume size in the medium can be efficiently measured by monitoring the variation in pick-off quenching lifetime. In the present study, we monitored the changes in *o*-Ps pick-off lifetime in ILs, both in the bulk and under confinement in SBA-15 nano-pores at different temperatures, to understand the temperature-dependent free-volume change, thereby gaining information on the molecular fragility. We correlated molecular fragility with the change in ionic self-diffusion inside the SBA-15 pores. The self-diffusivity and ionic conductivity of IL both in bulk and under confinement were studied using temperature-dependent broad-band dielectric spectroscopy (BDS) within a broad frequency range of 10^{−1}–10⁷ Hz. Thermogravimetric

analysis (TGA) and synchrotron-based SAXS were used to gain information on the degrees of pore filling with the ILs.

2. Materials and methods

1-Butyl-3-methylimidazolium-bis(trifluoromethylsulfonyl)-imide ([BmIm][NTf₂]) was procured from IoliTec, Germany, with purities of *ca.* 99% and was used without any further treatment. Acetone (LR grade) was purchased from Ruber-Jenson and used as such. Commercially available SBA-15 (Sigma Aldrich, with a specific pore volume of 1.39 cm³ g^{−1} and specific surface area of 1041 m² g^{−1}) was used to confine the IL. SBA-15 was preheated at 423 K in vacuum (10^{−3} mbar) for 4 hours to remove moisture.

3. Sample preparation

The IL + SBA-15 sample was prepared using the incipient technique for impregnation. A defined volume of IL (slightly higher than the total pore volume of blank SBA-15) was taken in a vial, and then, 10–15 mL of acetone was added and gently mixed to obtain a homogeneous mixture. Then, a pre-weighed amount of SBA-15 was added, and the mixture was shaken well overnight to ensure proper impregnation. The remaining IL molecules on the outer surface were removed by gentle washing with acetone. The composite was dried at 333 K overnight to remove any remaining acetone. Impregnated samples were stored in an airtight container to avoid exposure to moisture.

4. Experimental methods

4.1. X-ray diffraction (XRD) spectroscopy

X-ray diffraction patterns of samples were obtained at 2 θ values in the range of 0.5°–3° and 10°–40° using an X-ray powder diffractometer (proto AXRD benchtop powder diffractometer) with Cu K α = 1.54 Å as X-ray source.

4.2. FTIR spectroscopy

Fourier-transform infrared spectroscopy measurements were carried out in the wavenumber range of 500–4000 cm^{−1} using a Bruker Alpha II spectrometer in attenuated total reflection (ATR) mode.

4.3. Differential scanning calorimetry (DSC)

DSC measurements were carried out with a Mettler Toledo DSC (823e), Switzerland, at a scanning rate of 10 K min^{−1} in the temperature range of 170–290 K under argon atmosphere.

4.4. Thermogravimetric analysis (TGA)

The thermal stability of the matrix SBA-15, bulk IL and IL under confinement was analyzed using TGA (Setsys Evolution 1750, SETARAM, France) in the temperature range from room temperature (298 K) to 1073 K with a heating rate of 10 K min^{−1} under a flow of argon.

4.5. Positron annihilation lifetime spectroscopy (PALS)

A ^{22}Na positron source ($\sim 10 \mu\text{Ci}$) sealed within a thin Kapton foil ($5 \text{ mm} \times 5 \text{ mm} \times 8 \mu\text{m}$) was kept inside the powder sample in a hermetically sealed copper sample holder, which was then mounted on the cold head of a Sumitomo closed-cycle helium cryostat. Positron annihilation lifetime measurements were carried out using two BaF_2 scintillation detectors fitted with photomultiplier tubes (PMTs) connected in a fast-fast coincidence setup, which consists of a constant fraction differential discriminator (CFDD), ns delay, time to amplitude converter (TAC) and a multichannel analyzer (MCA). The energy windows of the two CFDDs were selected in such a way that one CFDD allows the energy pulse of the birth signal (1.276 MeV) of the positron, and the second allows the pulse of the death signal (0.511 MeV) of the positron coming from two respective PMT anode outputs. The timing pick-off signal from each CFDD is communicated to the TAC, and the positron annihilation lifetime histograms are finally collected in the MCA. The time resolution measured with ^{60}Co in a ^{22}Na window setting was 240 ps, and the time calibration was kept at 0.125 ns per channel. The spectrum with approximately 1×10^6 counts was acquired for each measurement, and data were analyzed using the PALSfit3 analysis program.²⁹ A fraction of positrons ($\sim 25\%$) annihilating in Kapton foil was taken into consideration as a source component (using a Si single-crystal as reference), which was subtracted from the PALS spectrum as a “source correction” while performing the data analysis. The nature of the PALS histogram is multi-exponential, which is represented in the following equation:

$$F(t) = \sum_{i=1}^k \frac{I_i}{\tau_i} e^{-t/\tau_i} \quad (1)$$

where $F(t)$ is the number of counts at time t , k is the number of exponential decay components, τ_i is the lifetime of the i th component having intensity I_i . The instrumental resolution (240 ps) is convoluted with the multi-exponential curves. The experimental spectra were best fitted into three or four (as the case may be) discrete lifetime components with the variance of fitting close to 1. The third (τ_3) and fourth (τ_4) lifetime components attributed to o -Ps lifetime in micro and mesopores, respectively, along with their corresponding intensities (I_1 and I_2 , respectively), are shown in Table 2. The o -Ps lifetime components were converted into corresponding average pore size using the following well-established Tao–Eldrup model:^{30,31}

$$\frac{1}{\tau_{\text{pick-off}}} = 2 \left[1 - \frac{R}{R + \Delta R} + \frac{1}{2\pi} \sin \left(\frac{2\pi R}{R + \Delta R} \right) \right] \quad (2)$$

where $\tau_{\text{pick-off}}$ (ns) and R (nm) are the o -Ps pick-off lifetime and corresponding pore radius, respectively. The parameter $\Delta R = 0.166 \text{ nm}$ is an empirical parameter that represents an electron layer thickness on the inner surface of the free volume or pore wall. With the increase in pore radius beyond $\sim 1 \text{ nm}$ (for mesopores), there is a chance of enhancement in intrinsic three-photon annihilation of o -Ps as well as annihilation from the higher excited states. In such a case, the extended Tao–Eldrup

model (ETE) is more suitable for converting o -Ps lifetime into mesopore radius.³²

Temperature-dependent PALS measurements were carried out for bulk IL and IL confined in SBA-15 nano-pores within the temperature range of 140–300 K with a rate of 1 K min^{-1} and an accuracy of $\pm 0.01 \text{ K}$. No hysteresis was observed for heating or cooling cycles, so the data presented here is of the cooling cycle.

4.6. Temperature-dependent broadband dielectric spectroscopy (BDS)

Broadband dielectric measurements on the bulk and confined IL were carried out at different temperatures (198–298 K) in a broad frequency range of 10^{-1} – 10^7 Hz using an impedance analyzer (Novocontrol, Alpha analyzer, Germany) coupled with a QuatroCryo system. The temperature stability during the measurement was maintained at $\pm 0.1 \text{ K}$ using a flow of dry N_2 . Circular powder samples with a diameter of 1.8 cm and a thickness of approximately 1 mm were used for the BDS measurements. The samples were positioned between standard circular gold-plated parallel electrodes, and an ac voltage of 0.2 V was applied to the electrodes. Measurements were carried out during both heating and cooling cycles, but no hysteresis was observed during the runs. Hence, only the cooling cycle data are presented in this article. BDS measures the complex dielectric function, $\epsilon^*(\omega) = \epsilon'(\omega) - i\epsilon''(\omega)$, which is equivalent to the complex conductivity function, $\sigma^*(\omega) = \sigma'(\omega) + i\sigma''(\omega)$, where ϵ' and ϵ'' are the real and imaginary parts of permittivity and σ' and σ'' are the real and imaginary parts of conductance, respectively. The complex conductivity and permittivity are related as $\sigma^*(\omega, T) = i\epsilon_0\omega\epsilon^*(\omega, T)$, indicating that $\sigma' = \epsilon_0\omega\epsilon''$ and $\sigma'' = \epsilon_0\omega\epsilon'$, where ϵ_0 is the vacuum permittivity, ω is the radial frequency ($= 2\pi f$) and T is the temperature. Modulus representation ($M^* =$

$1/\epsilon^* = i\omega\epsilon_0/\sigma^*$) with real $\left(M = \frac{\epsilon_0\omega\sigma''}{(\sigma')^2 + (\sigma'')^2} \right)$ and imaginary

$\left(M'' = \frac{\epsilon_0\omega\sigma'}{(\sigma')^2 + (\sigma'')^2} \right)$ parts gives the opportunity to take a

closer look at the ionic dynamics by suppressing dc conductivity and the electrode polarization contribution in the dielectric spectra.

4.7. Small-angle X-ray scattering (SAXS)

Small-angle X-ray scattering (SAXS) data of the samples were collected as a function of scattering vector q ($q = 4\pi \sin(\theta/2)/\lambda$, where θ is the scattering angle, and λ is the wavelength of probing radiation) in the range of 0.1 – 2.5 nm^{-1} . SAXS experiments were carried out at the beamline BL-18, Indus-2 synchrotron radiation facility (RRCAT, Indore, India).³³ This facility contains a 2D online image plate detector with a sample-to-detector distance of $\sim 3 \text{ m}$. For SAXS measurements, the powdered sample was packed in a circular slot, and both sides of the slot were covered with X-ray-transparent polyimide film. The SAXS data were corrected for background as well as empty

cell contributions. Fitting of these data was carried out using SASfit software.³⁴

4.8. Field-emission scanning electron microscopy (FESEM):

FESEM measurements on the SBA-15 before and after incorporation of IL were carried out using a GEMINISEM300 (Zeiss) at various magnifications.

4.9. N₂ gas adsorption isotherm

The surface area and pore size distribution of the blank SBA-15 and IL-loaded SBA-15 were obtained using the N₂ gas adsorption method (Sorptomatic 1990 surface area analyzer, CE Instruments, Italy) at 77 K. The samples were degassed at 473 K for 6 h before measurement.

5. Results and discussion

5.1. Characterizations

After impregnation of [BmIm][NTf₂] IL in the SBA-15 nanopores, the composite IL + SBA-15 was characterized using XRD, FTIR, N₂ gas adsorption isotherm, TGA, SAXS, FESEM and DSC techniques.

Fig. 1a and b show the small- and wide-angle XRD patterns, respectively. It can be seen in Fig. 1a that both SBA-15 and IL + SBA-15 show three characteristic peaks at $2\theta = 1.09^\circ$, 1.88° and 2.17° , corresponding to the (100), (110) and (200) planes, respectively, of the hexagonal mesoporous structure of SBA-15, which verifies that the structural framework of the matrix was maintained after incorporation of the IL.³⁵ In Fig. 1b, it is seen that there are broad peaks within the 2θ range of 15° to 35° corresponding to the amorphous structure (due to lack of long-range order) of SBA-15 particles.³⁶

Fig. 2 shows the N₂ adsorption-desorption isotherms for blank SBA-15. The pattern shows a type-IV physisorption isotherm with H2(b) hysteresis loops in the relative pressure range of 0.6 to 0.8, which is characteristic of ordered meso-structured materials.³⁷ The pore-size distribution was calculated using the Barrett-Joyner-Halenda (BJH) method, showing an average pore size of ~ 5 nm (dia) for blank SBA-15 (Fig. 2b). The values

for the pore volume and specific surface area are tabulated in Table 1. After incorporation of IL into the blank SBA-15, N₂ adsorption was found to be below the detectable limit, which is indicative of complete pore filling of the SBA-15 by the IL molecules, as corroborated by the SAXS and PALS measurements.

Fig. 3 presents the FTIR spectra of pristine SBA-15 (black), IL (red) and IL + SBA-15 (blue). The SBA-15 sample shows characteristic Si-O-Si stretching bands at around 801 cm^{-1} (symmetric) and 1064 cm^{-1} (asymmetric). Other notable features are a surface Si-OH asymmetric stretching band at 967 cm^{-1} , an Si-OH bending band at 1626 cm^{-1} and a broad band at around $3200\text{--}3600\text{ cm}^{-1}$, indicating a stretching band of surface -OH (see Fig. S1 in the SI for details).³⁸

The IL shows characteristic imidazolium hydrocarbon bands at around 3157 cm^{-1} and 3119 cm^{-1} and a shoulder to the last peak at 3102 cm^{-1} . Three alkyl C-H stretching bands can be seen at 2967 , 2941 and 2878 cm^{-1} . The peaks at 1176 (doublet) and 1227 cm^{-1} are assigned to the symmetric and asymmetric stretching bands of the -CF₃ group, respectively. In addition, peaks at around 1050 and 1347 cm^{-1} (doublet) correspond to the symmetric and asymmetric stretching bands of the -SO₃ group, respectively.³⁹ The bands at 569 , 740 and 789 cm^{-1} arise due to out-of-plane C-H rotation and bending of the imidazole ring. The peaks at $\sim 600\text{--}650\text{ cm}^{-1}$ are assigned to C-N-C skeletal bending of the imidazolium ring. The broad band at $\sim 843\text{ cm}^{-1}$ corresponds to in-plane imidazole ring bending.⁴⁰ Upon confinement of the IL into the nano-pores of SBA-15, the composite carries the vibrational signature of both the matrix and IL (indicated by arrows). Under confinement, the positions of vibrational peaks or bands do not change, indicating that the interaction between the IL molecules and the surface of the SBA-15 is not significant. But, notably, upon confinement, the vibrational bands of the IL are broadened substantially due to a curtailing of the cooperative length of the long IL molecular chain, as well as a difference in the local environment between the IL molecule near the pore wall surface and others at the core of the pore, resulting in a broader distribution of vibrational frequencies.

The PAL spectrum for blank SBA-15 was fitted with four lifetime components (Fig. S2 in SI). The two long-lived *o*-Ps

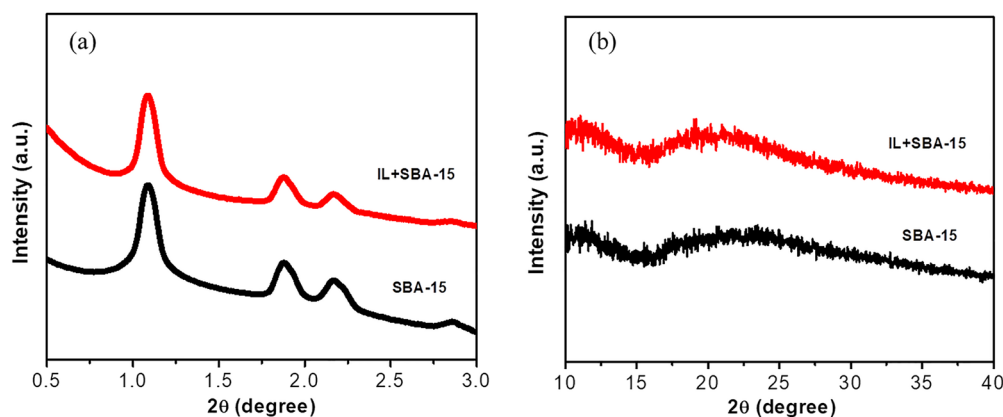


Fig. 1 (a) Small-angle and (b) wide-angle XRD patterns of the blank SBA-15 matrix and IL + SBA-15 composite.

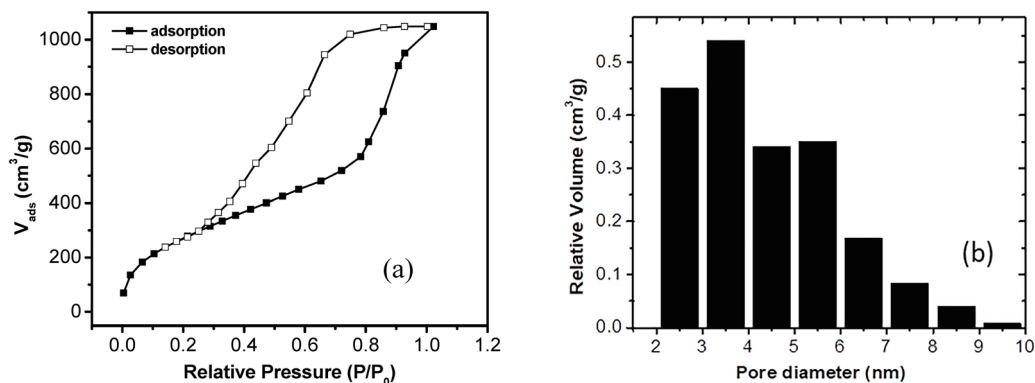


Fig. 2 (a) N_2 adsorption-desorption isotherm and (b) pore diameter distribution of blank SBA-15.

Table 1 Specific pore surface area and pore volume calculated from BET N_2 gas adsorption-desorption data

Sample	Specific pore surface area ($m^2 g^{-1}$)	Specific pore volume ($cm^3 g^{-1}$)
SBA-15	1041	1.39
IL + SBA-15	—	—

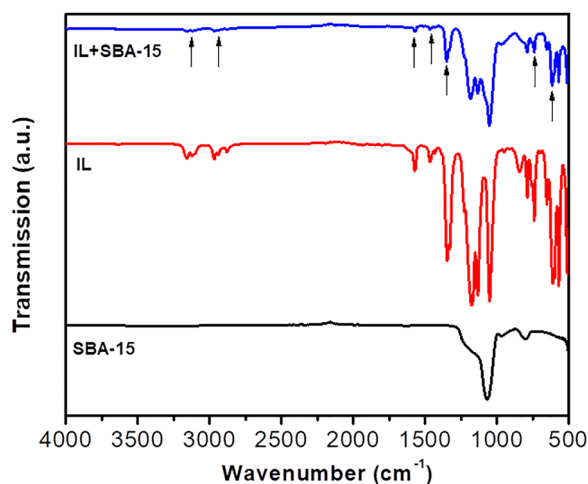


Fig. 3 FTIR spectra of the blank SBA-15, IL and IL + SBA-15 composite.

lifetime components (τ_3 and τ_4) and their corresponding intensities (I_3 and I_4) in blank SBA-15 are shown in Table 2. In the empty pore, the shorter component (τ_3 and I_3) is attributed to o -Ps pick-off annihilation in the micropores present at the surface of the mesopore walls. On the other hand, the longer component (τ_4 and I_4) corresponds to o -Ps annihilation inside the mesopores, as well as in the inter-granular spaces after diffusing out from the mesopores. It is interesting to observe

Table 2 Measured o -Ps lifetimes and their intensities

Sample	τ_3 (ns)	I_3 (%)	τ_4 (ns)	I_4 (%)
SBA-15	3.59 ± 0.15	2.87 ± 0.13	129.77 ± 1.55	10.61 ± 0.07
IL	3.45 ± 0.02	14.17 ± 0.08	—	—
IL + SBA-15	3.58 ± 0.02	17.16 ± 0.13	—	—

that once the pores are completely filled with IL, the τ_4 component completely disappeared, leaving only the third component (τ_3), confirming the filling of the mesopores with the IL. The total o -Ps intensity contribution ($\sim 17\%$) for the IL incorporated SBA-15 system comes out to be exactly the sum of o -Ps intensity for blank SBA-15 micropores and IL in bulk. The micropore size in the blank SBA-15 (~ 0.8 nm) was calculated from τ_3 using the Tao-Eldrup model³² (Table 3). The value of τ_4 (129 ns), as seen in Table 2, is the mean value of o -Ps lifetime in the mesopore and intergranular spaces. Hence, this component cannot give us an exact value of the mesopore size. Therefore, this component was not used to calculate the mesopore size. The mesopore-size distribution was obtained from the Brunauer-Emmett-Teller (BET) N_2 gas adsorption/desorption isotherm data (Fig. 2), which shows that the average mesopore size is around 5 nm. Using SAXS data, the calculated distance between two adjacent mesopore centers is around 11 nm (detailed calculations are described in the SI), which is seen to be higher than the mesopore diameter obtained from the BET data. The higher value of the inter-pore distance as compared to the average pore diameter is due to the presence of the thick pore-wall structure in SBA-15. It is reported that the average pore-wall thickness in another silica SBA group (SBA-16) is around 5.3 nm.⁴¹ If this value of wall thickness is considered for SBA-15, then the pore diameter comes to around 5.7 nm, which is still slightly higher than the value obtained from the BET isotherm. The higher value may be due to our consideration of a sharp pore boundary instead of a diffused boundary with higher average pore-wall density ($2.2 g cm^{-3}$, the same as the amorphous silica density)⁴²

The fraction of pore filling with IL in the SBA-15 matrix was quantified using thermo-gravimetric (TGA) analysis, as shown in Fig. 4, which shows the wt% loss of pristine SBA-15, bulk IL and IL + SBA-15 at high temperatures. The pristine SBA-15 does not show any significant mass loss, except for a slight decrease

Table 3 Micropore sizes

Sample	Micropore size (diameter) (nm)
SBA-15	0.795 ± 0.018
IL + SBA-15	0.793 ± 0.003

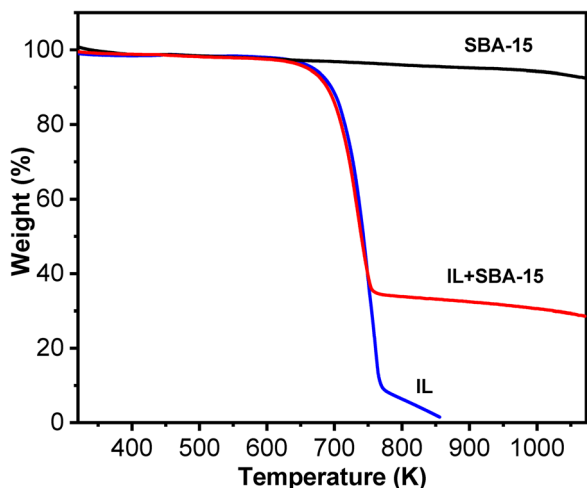


Fig. 4 Thermogravimetric curves for blank SBA-15, pure IL and the IL + SBA-15 composite obtained at a heating rate of 10 K min⁻¹.

in mass up to 403 K, indicating a continuous loss of adsorbed water in small quantities from the matrix. On the other hand, it is seen that the pure IL starts to decompose at around 630 K. Interestingly, the TGA profile of the IL + SBA-15 composite does not show much difference from that of pure IL, probably due to the reduced interaction of the IL with the pore wall due to the hydrophobicity of the bulky anion NTF₂⁻. The fraction of pore filling is estimated from the TGA after deducting the contribution of water loss from the pristine SBA-15. The fraction of pore-filling comes out to be 65.0% (w/w), which is equivalent to around 92.8 vol%, considering the density of IL (1.44 g cm⁻³) and the specific pore volume of SBA-15 (1.39 cm³ g⁻¹).

SAXS data (Fig. 5) also provides information about the pore-filling fraction of SBA-15 with IL. Fig. 5 shows characteristic peaks corresponding to the (100), (110) and (200) planes, common in the 2D hexagonal space group *P6mm*.⁴³ It is seen that the peak positions are not shifted after IL loading inside

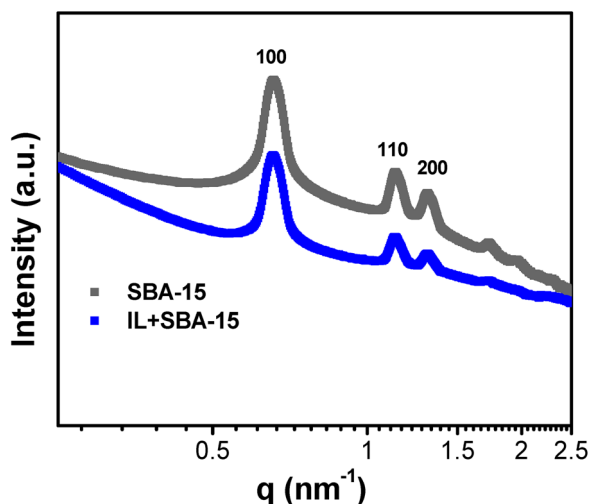


Fig. 5 SAXS pattern of the blank SBA-15 and IL + SBA-15 samples, showing the decrease in the intensity of the (100) peak after pore filling.

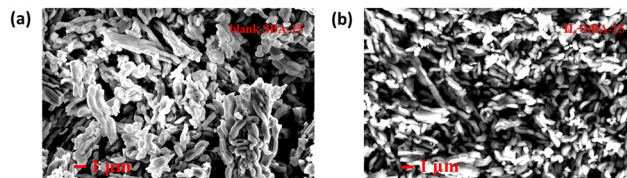


Fig. 6 FESEM images of the (a) blank SBA-15 and (b) IL + SBA-15 samples.

the pores, confirming that the unit cell parameters are unchanged. The intensity of the scattering peaks in the IL + SBA-15 composite is seen to be decreased compared to that in empty SBA-15, pointing towards the fact that the mesopores of the SBA-15 moiety are filled by the IL, leading to a decrease in contrast and a decrease in the intensity. The pore filling fraction was calculated from the intensity information of the (100) peaks of the blank SBA-15 and IL + SBA-15 samples (detailed in the SI). The calculated value of the pore-filling fraction is around 93 vol%, which agrees well with the value calculated from TGA analysis.

The morphology and microstructure of SBA-15 and IL + SBA-15 grains were investigated using FESEM. As seen from the FESEM images (Fig. 6), the microstructure of the SBA-15 host remains unchanged upon incorporation of the ionic liquid into the pores. Further magnified FESEM images for both blank SBA-15 and IL + SBA-15 are shown in Fig. S3 in the SI.

Fig. S4 in the SI shows the DSC profiles of the IL both in the bulk and confined state in SBA-15 nano-pores during the heating cycle. The sample was first cooled to 170 K and then slowly heated at a rate of 10 K min⁻¹. During heating, a step-like signature in the DSC profile at around 186 K for the bulk IL corresponds to a change from glassy to rubbery state. Subsequently, with a further increase in temperature, the mobility of the rubbery phase increases, and a cold crystallization can be observed at a temperature (*T_{cc}*) of around 237 K, followed by a melting peak at a higher temperature (*T_m*) of around 272 K. The exo- or endo-thermic peaks in the DSC profile (Fig. S4) were observed to become sharper on going from the bulk to confined IL state, possibly due to a breakdown of the long-range order of the IL under confinement. The glass transition temperature (*T_g*), cold crystallization temperature (*T_{cc}*) and melting temperature (*T_m*) for both the bulk and confined IL samples are tabulated in Table 4. An expanded view of the DSC profile is shown in Fig. 7, where the glass transition temperature (*T_g*) of the IL under both conditions (bulk and confinement) is clearly visible.

5.2. Temperature-dependent PALS study

The temperature dependence of the *o*-Ps pick-off lifetime (τ_3) in bulk and confined IL within a temperature range of 140–300 K

Table 4 Transition temperatures obtained from the DSC curves

Sample	<i>T_g</i> (K)	<i>T_{cc}</i> (K)	<i>T_m</i> (K)
IL	186	237	272
IL + SBA-15	184	250	260

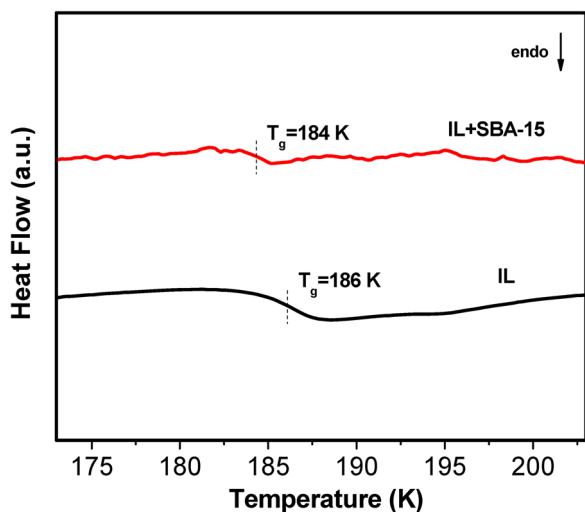


Fig. 7 Glass transition temperatures (T_g) of bulk IL and confined IL (IL + SBA-15) marked in the DSC profiles.

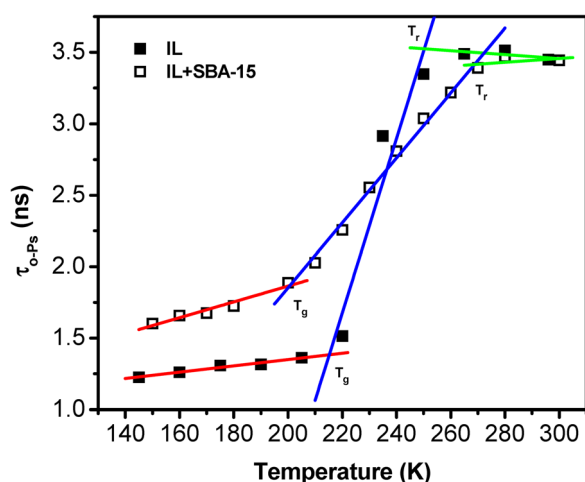


Fig. 8 Temperature dependence of *o*-Ps pick-off lifetime (τ_3) for IL in bulk and under confinement (experimental errors are smaller than the size of the symbols).

is shown in Fig. 8. It is seen that the temperature profile of τ_3 is qualitatively similar to other molecular glass formers reported in the literature.^{44,45,51,52} The characteristic temperatures can be identified from the thermal response of the τ_3 for both bulk and confined IL. At the lower temperature range (below 200 K), the τ_3 value shows weak temperature dependence, followed by a much stronger temperature dependence at an intermediate temperature range (~ 200 – 270 K), with a crossover at the glass transition temperature (T_g) at 215 K for bulk and 201 K for confined IL. A second crossover is seen at a higher temperature of $T_r = 250$ K for bulk and 269 K for the confined IL, above which τ_3 slowly varies with temperature. At lower temperatures (below T_g), the molecules are frozen into the glassy state, and thus, τ_3 reflects the static free volume in the molecular network.⁴⁸ With the increase in temperature above T_g , the mobility of the molecules increases rapidly, resulting in a

strong temperature dependence for the τ_3 component, pointing towards a rapid increase in the local free volume until the second crossover temperature T_r .^{48–50} The values of T_g and T_r are extracted from the intercept of a straight-line fitting of *o*-Ps pick-off lifetime data (Fig. 8).⁴⁷ The T_g and T_r can also be estimated from the temperature variation of the *o*-Ps intensity (I_3) plot (see Fig. S5 in the SI). Table 5 summarizes the values of T_g and T_r for IL in bulk and in confinement. The T_g value determined from the DSC differs from that estimated from PALS because of the possible difference in the probing aspect; *i.e.*, DSC looks at bulk thermal transition, whereas PALS detects changes in the free volume at the molecular level. It is interesting to note that the changes in the value of τ_3 within the temperature range between T_g and T_r are sharper in the case of bulk IL than in the case of confined IL. Additionally, the difference between T_g and T_r is seen to be smaller for bulk IL than for confined IL [$(T_r/T_g)_{\text{bulk IL}} < (T_r/T_g)_{\text{confined IL}}$]. In case of bulk IL, the sharp change in τ_3 within the narrow temperature range of T_g and T_r suggests a pronounced thermal transition due to inefficient molecular packing or high configurational freedom (fragile glass former), with a high degree of structural inhomogeneity, resulting in a large free-volume change during the glass transition.⁴⁶ In confined IL, on the other hand, the smaller changes in the value of τ_3 within the broader temperature window of T_g and T_r compared to that of bulk IL suggest a smaller free-volume change within the broad range of T_g and T_r . This implies that the IL has a relatively compact molecular packing compared to its fragile-glass-forming state in the bulk, thus showing much broader transitions in confinement.^{46,51} The smaller free-volume change for the IL in a wider temperature range under nano-confinement infers that the fragility of the IL is reduced while confined in nano-pores.⁴⁶ The same conclusion can also be drawn from the intensity plot of the *o*-Ps pick-off component (I_3) (see Fig. S5 in the SI). The higher T_r/T_g value for confined IL (~ 1.34) than that for the bulk (~ 1.16) is in line with the reported values for less to more fragile glass formers, respectively (Table 5).⁴⁶ In a fragile glass former liquid, the change in free volume during thermal transition depends on the molecular mobility.⁵¹ Bulk IL has a wide range of molecular motions and long-range cooperative relaxation having spatial freedom, thus showing a large free-volume change during the glass transition; however, under nano-confinement, spatial restriction limits the cooperative dynamics and molecular motions, resulting in a reduction in the fragility of the glass-forming liquid. From both the DSC and PALS studies, it is noted that, under nano-confinement, there is a slight reduction in the T_g of the IL from its bulk value (~ 2 K in DSC and ~ 14 K in PALS). Weak interactions of the cation (BmIm^+) and anion (NTf_2^-) with the pore wall (silanol group) lead to a depression

Table 5 Parameters obtained from temperature-dependent PALS study

Sample	T_g (K)	T_r (K)	T_r/T_g
IL	215	250	1.16
IL + SBA-15	201	269	1.34

in the glass transition temperature for nano-confined IL, as evident from the experimental data.⁵² The high-temperature crossover (T_r) above the glass transition may have different origins. There are many experimental studies in which possible reasons behind the appearance of the crossover temperature (T_r) have been discussed, for example, decoupling of rotational and translational motion of supercooled liquid near the T_r , resulting in failure of the Stokes–Einstein relation on diffusion and viscosity, and Debye–Einstein relation on reorientation time and viscosity below T_r ,^{53–55} decoupling of α and β relaxation processes,^{56,57} the breaking of ergodic nature as predicted by mode-coupling theory,^{58,59} and a crossover between two VFT-like behaviors above the glass transition temperature.^{60,61}

5.3. Dielectric characterization

The transformation of fragility of the glass-forming IL from high to low value while confined in nano-pores of SBA-15 was further explored using broadband dielectric spectroscopy (BDS) in the broad frequency range of 10^{-1} – 10^7 Hz to characterize the ion dynamics under nano confinement. The ion conductivity and the diffusion coefficient of ions in the IL in the bulk and in the nano-pores of SBA-15 were probed within the temperature range of 198–298 K. The temperature-dependent real part (σ') of the complex conductivity (σ^*) and the imaginary part (M'') of the complex dielectric modulus (M^*) within the broad frequency range of 10^{-1} – 10^7 Hz are shown in Fig. 9(a)–(d). The real (ϵ') and imaginary (ϵ'') parts of the complex permittivity

(ϵ^*) of the bulk and confined IL within the temperature range of 198–298 K are shown in Fig. S6(a)–(d). The rising of both $\epsilon'(f)$ and $\epsilon''(f)$ at low frequency range and subsequently reaching at a plateau region at further lower frequency region is due to electrode polarization⁶² which is very common for blocking of electrodes that do not allow the conducting ions to penetrate. This phenomenon is also seen as a decrease in the conductivity at lower frequencies in the $\sigma'(f)$ plot (Fig. 9a and c). We can assign a characteristic frequency (f_c) in the $\sigma'(f)$ profile (Fig. 9a and c), above which the deviation from dc conductivity (σ_{dc}) sets in and $\sigma'(f)$ follows the Jonscher's power law (see eqn (3)).^{63,64} This characteristic frequency (f_c) can be identified as the frequency at which the imaginary part (M'') of the dielectric modulus shows a peak (shown in Fig. 9b and d).^{65–67}

Fig. S7(a) and (b) show the fitting of the real part of the ionic conductivity (σ') at different temperatures. At frequency ranges above f_c , the conductivity follows Jonscher's power law given in the following equation:⁶²

$$\sigma'(f) = \sigma_{dc} + Af^n \quad (3)$$

where A is a constant that dictates polarization strength and parameter n dictates the interaction between the ionic species and their surrounding environment.⁶⁸ The fitting results are summarized in Tables S2 and S3 in the SI for bulk and confined IL, respectively. The values of n generally remain within 0.6 to 1 for most of the ionic compounds,⁶⁹ and the transition in value of n from <1 to >1 suggests a change in the conduction

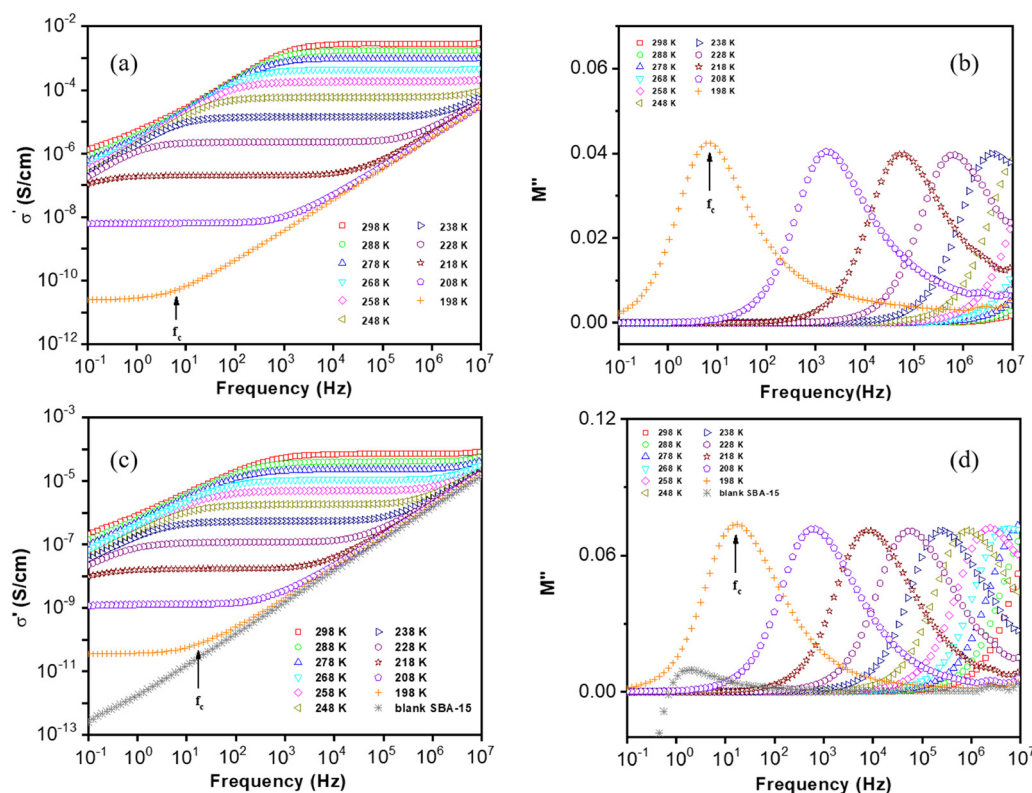


Fig. 9 Dielectric spectra of bulk IL [(a) and (b)] and confined IL in SBA-15 [(c) and (d)]. (a) and (c) show the real part (σ') of the complex conductivity, and (b) and (d) show the imaginary part (M'') of the dielectric modulus. Data for blank SBA-15 are shown at 258 K temperature.

mechanism of charge carriers (ions) from hopping to quantum tunneling mode.^{70,71} It is to be noted that the origin of Jonscher's power law lies in the relaxation of the mobile ionic species in the medium.⁷² However, a small intrinsic relaxation process, as observed at high frequencies and low temperatures in ϵ' and ϵ'' profile (Fig. S6), which may correspond to the re-orientational dipole motion as proposed by Gainaru *et al.*,⁷³ can also be responsible for the variation of σ' with frequency in the present perspective.^{74,75} Hence, the actual origin of Jonscher's power law in the present case may remain unresolved.

The M'' spectra (Fig. 9b and d) are commonly utilized to determine the conductivity relaxation time (τ), which is indicative of the characteristic timescale of ionic dynamics. We have employed a formalism to calculate the relaxation time ($\tau = 1/2\pi f_c$) by using the main peak in the M'' spectra, which characterizes the time scale of the dynamics of the IL.⁶⁶ With the decrease in temperature, the peaks in the M'' spectra shift to lower frequency due to a slowing down of the ionic dynamics as the viscosity of the medium increases.⁷⁴ We have calculated conductivity-relaxation times at each temperature for IL, both in bulk and confined inside SBA-15 nano-pores. Upon scaling the $\sigma'(f)$ and $\sigma''(f)$ with respect to f_c and σ_c (value of $\sigma'(f)$ at f_c) over the whole temperature range, the spectra for both bulk IL and IL + SBA-15 superimpose, indicating identical temperature dependence of transport and electrode blocking properties both in bulk and in confinement (see Fig. S8).¹⁷ As the conductivity-relaxation peak in the M'' spectra above 258 K goes out of the frequency range of BDS measurements for both IL and IL + SBA-15, we use σ_{dc} data (as obtained from fitting of eqn (3)) for further analysis and correlation. Fig. 10a and b show the variation of the dc conductivity (σ_{dc}) and the relaxation time (τ), respectively, with temperature. Both $\log(\sigma_{dc})$ and $\log(\tau)$ show non-linear behavior with temperature, indicating deviation from the Arrhenius law throughout the temperature range. This type of deviation from the simple activation process (Arrhenius behavior) for IL's has been reported earlier.^{76–79} The temperature dependence of σ_{dc} and τ for IL and the IL + SBA-15 composite was fitted with the Vogel-Fulcher-Tammann (VFT) equation as follows:^{80,81}

$$\tau = \tau_0 \times \exp\left(\frac{B}{T - T_0}\right) \quad \text{and} \quad \sigma_{dc} = \sigma_0 \times \exp\left(-\frac{B}{T - T_0}\right) \quad (4)$$

where τ_0 (inverse attempt frequency) and σ_0 are pre-exponential parameters, B is the fragility parameter, and T_0 is known as the Vogel temperature. The best fitted values of B and T_0 for both IL and IL + SBA-15 are tabulated in Table S1 in the SI. The fragility parameter B can be replaced with DT_0 , where D is known as the strength parameter of the glass-forming IL and can be used to determine the degrees of fragility of the IL through the following relation:^{76,82,83}

$$m = 16 + \frac{590}{D} \quad (5)$$

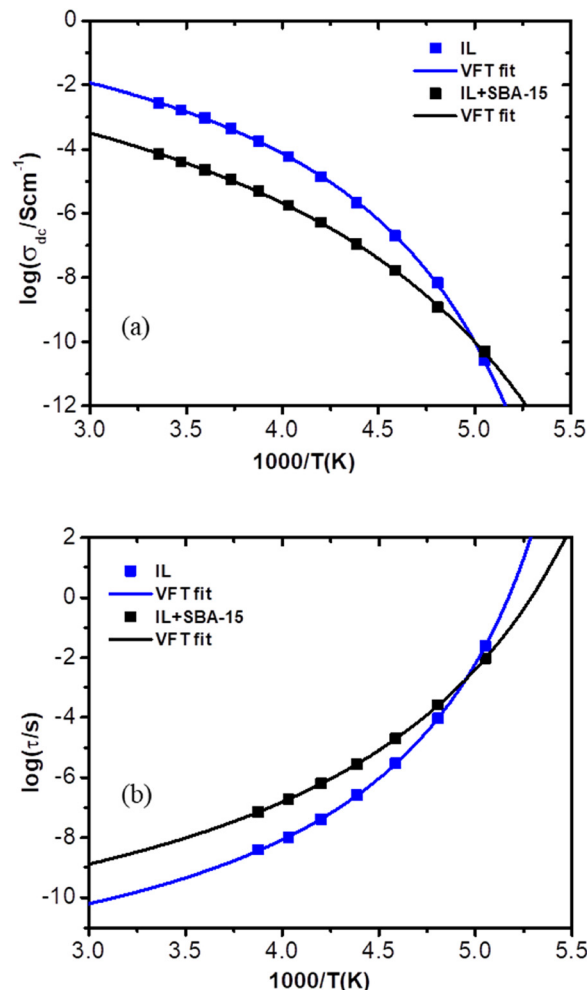


Fig. 10 Variation in the (a) dc conductivities and (b) relaxation times of IL and IL + SBA-15 with temperature. The values are determined from the fittings of the dielectric spectra (M'' and σ') at different temperatures. The solid lines are the VFT fittings of the dc conductivity and relaxation time.

Table 6 Values of fragility parameter (m) obtained using eqn (5). The parameter D is obtained from the VFT fittings of τ and σ_{dc}

	m (IL)	m (IL + SBA-15)
From relaxation (τ)	125 ± 4	96 ± 2
From dc conductivity (σ_{dc})	119 ± 2	83 ± 3

The values of the fragility index m , obtained from the best fitted value of B for both σ_{dc} and τ using the VFT fitting equation in IL and IL + SBA-15, are shown in Table 6.

The lower value of the fragility index m in the case of IL confined in SBA-15 nano-pores compared to that of bulk IL, estimated from both σ_{dc} and τ , corresponds to the stronger (less fragile) glass former under nano-confinement,⁸⁴ supporting the PALS result obtained from the *o*-Ps pick-off lifetime profile with temperature. The glass transition point (T_g) of the ionic subsystem can be identified as the temperature at which the relaxation time τ equals 100 s and $\sigma_{dc} = 10^{-14}$ S cm⁻¹ (in a few reports, $\sigma_{dc} = 10^{-15}$ S cm⁻¹).^{85–87} The T_g values obtained

Table 7 Value of T_g (K) obtained from VFT fittings

	T_g (K) (IL)	T_g (K) (IL + SBA-15)
From relaxation time (at $\tau = 100$ s)	189	183
From dc conductivity (at $\sigma_{dc} = 10^{-14}$ S cm $^{-1}$)	189	182
From dc conductivity (at $\sigma_{dc} = 10^{-15}$ S cm $^{-1}$)	187	180

from the extrapolated plots of both τ and σ_{dc} for bulk IL and confined IL are given in Table 7 (more clear pictures of both the plots are given in Fig. S9a and b in the SI).

As seen in Table 7, the value of T_g obtained from both the extrapolated data plots is lower in the case of confined IL compared to that in bulk IL and is within the range of T_g values obtained from the DSC measurements.

To further explore the ionic conduction mechanism of the IL in the bulk and confinement, the ionic diffusion coefficient (D) of ions and its manifestation on the conduction behavior of IL were analyzed. The diffusion coefficient (D) can be extracted from the BDS spectra by a combination of Einstein and Einstein-Smoluchowski equations:⁸⁸

$$D = \frac{\lambda^2 \omega_c}{2} \quad (6)$$

where λ is the mean jump length of ions on the timescale of ω_c ($2\pi f_c$, characteristic frequency at which dispersion sets in for dc conductivity, as mentioned earlier, and also indicated in the plot in the SI). The value of λ is taken as 0.26 nm⁸⁹ for the calculation of D at different temperatures. The variation of D with temperature is shown in Fig. S10, which follows the same trend as the dc conductivity (σ_{dc}), as expected.

It is interesting to note that the value of σ_{dc} (or D) is less for IL confined in nano-pores of SBA-15 compared to that in the bulk at room temperature, which indicates that confinement restricts the mobility of the charge carriers. This result is also supported by the reported observation of the decrease in conductivity with the decrease in fragility for ionic liquids.⁶⁶ Inside the confinement, the IL becomes a less fragile glass-forming liquid, with a rigid structure making it more difficult for the ions to diffuse. However, with the decrease in temperature, the rate of decrease of σ_{dc} (or D) in confined IL is seen to be less than that in the bulk, which may be due to the restriction in the growth of cooperative length inside the pore, resulting in lesser viscous force as compared to that in bulk IL at low temperature. With a further decrease in temperature, the value of σ_{dc} (or D) for the confined IL approaches the value of that in the bulk, and at temperatures around 200 K, the σ_{dc} (or D) in the confined IL exceeds its corresponding value in the bulk IL, indicating faster ion mobility in confined IL compared to that in bulk below the glass transition temperature (T_g). Below T_g , the molecular free volume size in the confined IL is seen to be larger than that in the bulk IL, which has a noticeably weaker temperature dependence, as shown in Fig. 8. The cooperativity length scale of confined glassy IL may grow to the limit of the confining pore size below T_g . The combined effect of the larger free volume and the limited

cooperative length of the IL may enhance the ionic conductivity inside the nano-pores below T_g . However, we could not extract the σ_{dc} value from the present BDS curve below 198 K because, beyond ~ 200 K (*i.e.*, as the temperature approaches T_g), the ionic mobility slows down dramatically, shifting the conductivity-relaxation time to a much longer time scale; thus, it goes out of the range of the current experimental frequency window at further lower frequency. In future studies, it will be of high interest to perform measurements well below 200 K to observe the exact profile of σ_{dc} in confinement as compared to that in the bulk over a broader frequency range. Further work on the nanotribology of [BmIm][NTf₂] inside the nano-pores may give a more accurate explanation for the difference in the temperature profile of the diffusion coefficient in bulk and under nano-confinement.

6. Conclusion

In the present study, [BmIm][NTf₂] IL was incorporated inside the nano-pores of the SBA-15 crystalline mesoporous silica matrix. We demonstrated that the structural transformation of the glass-forming liquid from high to low fragility and the glass transition dynamics of confined [BmIm][NTf₂] IL can be accurately monitored through the temperature-dependent variation in the local free volume of the IL using PALS, which is supported by the results obtained through BDS measurements. The BDS analysis further showed that the [BmIm][NTf₂] IL becomes a stronger (less fragile) glass former upon confinement into SBA-15 nanopores affecting the diffusion of ions at room temperature. It is observed that at low temperatures (<200 K), the ionic conductivity and diffusion coefficient of ions inside the nano-confinement supersede their corresponding values in the bulk, which gives a strong indication of the enhancement in ionic conductivity inside the nano-confinement below the glass transition temperature compared to that in the bulk. Our findings thus have a fundamental importance in describing the low-temperature behavior of the IL inside a nano-confinement. Further studies on a broader frequency range at low temperatures will be of high interest to gain more detailed information about the enhanced conductivity of the [BmIm][NTf₂] ions under nano-confinement.

Conflicts of interest

.There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information: additional calculations, graphs of experimental data and tables of fitted values. See DOI: <https://doi.org/10.1039/d5cp03463a>.

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